



Department of Computer Science and Engineering (Data Science)

Subject: Machine Learning – I (DJS23DPC252)

AY: 2025-26

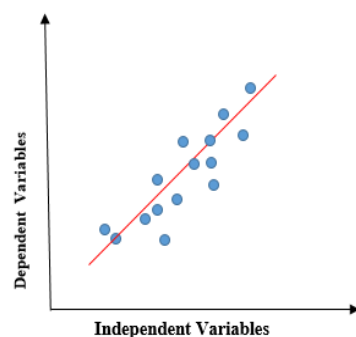
Experiment 1

(Regression)

Aim: Implement Linear Regression on the given Dataset and apply Regularization to overcome overfitting in the model.

Theory:

- **Linear Regression:** Linear regression is a quiet and simple statistical regression method used for predictive analysis and shows the relationship between the continuous variables. Linear regression shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis), consequently called linear regression. *If there is a single input variable (x), such linear regression is called **simple linear regression**. And if there is more than one input variable, such linear regression is called **multiple linear regression**.* The linear regression model gives a sloped straight line describing the relationship within the variables.



The above graph presents the linear relationship between the dependent variable and independent variables. When the value of x (**independent variable**) increases, the value of y (**dependent variable**) is likewise increasing. The red line is referred to as the best fit straight line. Based on the given data points, we try to plot a line that models the points the best. *To calculate best-fit line linear regression uses a traditional slope-intercept form.*



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$$y = mx+b \implies y = a_0 + a_1x$$

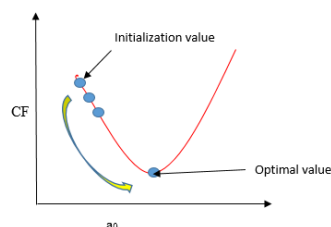
y= Dependent Variable; x= Independent Variable; a0= intercept; a1 = Linear regression coefficient.

- **Cost function:** The cost function helps to figure out the best possible values for a0 and a1, which provides the best fit line for the data points. Cost function optimizes the regression coefficients or weights and measures how a linear regression model is performing. The cost function is used to find the accuracy of the **mapping function** that maps the input variable to the output variable. This mapping function is also known as **the Hypothesis function**. In Linear Regression, **Mean Squared Error (MSE)** cost function is used, which is the average of squared error that occurred between the predicted values and actual values. *By simple linear equation $y=mx+b$ we can calculate MSE as: Let's y = actual values, y_i = predicted values*

$$MSE = \frac{1}{N} \sum_{i=1}^n (y_i - (mx_i + b))^2$$

Using the MSE function, we will change the values of a0 and a1 such that the MSE value settles at the minima. Model parameters **$x_i, b (a_0, a_1)$** can be manipulated to minimize the cost function. These parameters can be determined using the gradient descent method so that the cost function value is minimum.

- **Gradient descent:** Gradient descent is a method of updating a0 and a1 to minimize the cost function (MSE). A regression model uses gradient descent to update the coefficients of the line ($a_0, a_1 \Rightarrow x_i, b$) by reducing the cost function by a random selection of coefficient values and then iteratively update the values to reach the minimum cost function.





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To update a_0 and a_1 , we take gradients from the cost function. To find these gradients, we take partial derivatives for a_0 and a_1 .

$$J = \frac{1}{n} \sum_{i=1}^n (a_0 + a_1 \cdot x_i - y_i)^2$$

$$\frac{\partial J}{\partial a_0} = \frac{2}{n} \sum_{i=1}^n (a_0 + a_1 \cdot x_i - y_i)$$

$$\frac{\partial J}{\partial a_1} = \frac{2}{n} \sum_{i=1}^n (a_0 + a_1 \cdot x_i - y_i) \cdot x_i$$

$$\frac{\partial J}{\partial a_0} = \frac{2}{n} \sum_{i=1}^n (pred_i - y_i)$$

$$\frac{\partial J}{\partial a_1} = \frac{2}{n} \sum_{i=1}^n (pred_i - y_i) \cdot x_i$$

$$a_0 = a_0 - \alpha \cdot \frac{2}{n} \sum_{i=1}^n (pred_i - y_i)$$

$$a_1 = a_1 - \alpha \cdot \frac{2}{n} \sum_{i=1}^n (pred_i - y_i) \cdot x_i$$

Partial derivatives are the gradients and they are used to update the parameters.

- **Regularization:** When linear regression is underfitting there is no other way (given you can't add more data) then to increase complexity of the model making it polynomial regression (cubic, quadratic, etc...) or using other complex model to capture data that linear regression cannot capture due to its simplicity. When linear regression is overfitting, number of columns (independent variables) approach number of observations there are two ways to mitigate it
 1. Add more observations
 2. Regularization

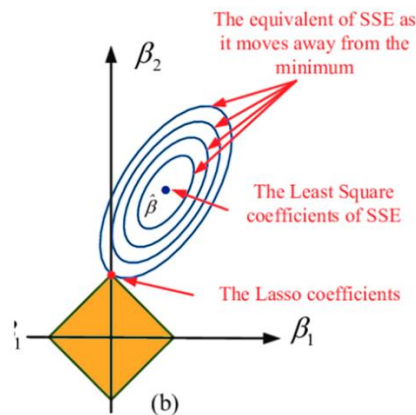
Since adding more observations is time consuming and often not provided we will use regularization technique to mitigate overfitting. There are multiple regularization techniques, all

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share the same concept of **adding constraints on weights** of independent variables(except θ_0) however they differ in way of constraining. We will go through three most popular regularization techniques: Ridge regression (L2) and Lasso regression (L1)

- **Lasso Regression**

The word "LASSO" denotes Least Absolute Shrinkage and Selection Operator. Lasso regression follows the regularization technique to create prediction. It is given more priority over the other regression methods because it gives an accurate prediction. Lasso regression model uses shrinkage technique. In this technique, the data values are shrunk towards a central point similar to the concept of mean. The lasso regression algorithm suggests a simple, sparse models (i.e. models with fewer parameters), which is well-suited for models or data showing high levels of multicollinearity or when we would like to automate certain parts of model selection, like variable selection or parameter elimination using feature engineering. Lasso Regression algorithm utilises L1 regularization technique It is taken into consideration when there are more number of features because it automatically performs feature selection.



Residual Sum of Squares + λ * (Sum of the absolute value of the coefficients)

The equation looks like:

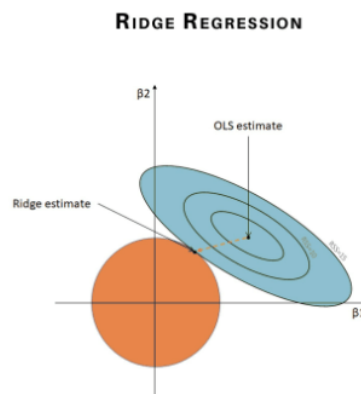
$$\sum_{i=1}^n (y_i - \sum_j x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j|$$



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- **Ridge Regression**

Ridge Regression is another type of regression algorithm in data science and is usually considered when there is a high correlation between the independent variables or model parameters. As the value of correlation increases the least square estimates evaluates unbiased values. But if the collinearity in the dataset is very high, there can be some bias value. Therefore, we create a bias matrix in the equation of Ridge Regression algorithm. It is a useful regression method in which the model is less susceptible to overfitting and hence the model works well even if the dataset is very small.



The cost function for ridge regression algorithm is:

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Where λ is the penalty variable. λ given here is denoted by an alpha parameter in the ridge function. Hence, by changing the values of alpha, we are controlling the penalty term. Greater the values of alpha, the higher is the penalty and therefore the magnitude of the coefficients is reduced. We can conclude that it shrinks the parameters. Therefore, it is used to prevent multicollinearity, it also reduces the model complexity by shrinking the coefficient.



Shri Vile Parle Kelavani Mandal's

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(Autonomous College Affiliated to the University of Mumbai)

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Lab Assignments to complete in this session

1. Create synthetic dataset with one variable, divide dataset into 70% train and 30% split, apply linear regression using library and evaluate prediction results using Mean Square Error (MSE).
2. Create synthetic dataset with multiple variables, divide dataset into 70% train and 30% split, apply linear regression using library and evaluate prediction results using Mean Square Error (MSE).
3. Implement manual gradient descent and manual stochastic gradient descent on the above dataset and plot the training loss curve.
4. Apply Regression, Lasso and Ridge regression on "Diabetes.csv" dataset from the library and compare Mean Square Error (MSE).
5. Apply Regression, Lasso and Ridge regression on "Boston.csv" dataset and compare Mean Square Error (MSE).